MySQL 8.0 Command Line Cli

```
mysql> -- Q1: Write a query to find the addresses (location_id, street_address, city, state_province, country_name) of all the departments.
mysql> select location_id, street_address, city, state_province, country_name from departments natural join locations natural join countries;
+-------------+----------------+-------------------+----------------+--------------------------+
| location_id | street_address | city              | state_province | country_name             |
+-------------+----------------+-------------------+----------------+--------------------------+
|        1400 | 12 Lakeview Dr | Southlake         | Texas          | United States of America |
|        1500 | 45 Bay Street  | South San Francisco | California   | United States of America |
|        1700 | 300 Pine Avenue | Seattle          | Washington     | United States of America |
|        1700 | 300 Pine Avenue | Seattle          | Washington     | United States of America |
|        1700 | 300 Pine Avenue | Seattle          | Washington     | United States of America |
|        1700 | 300 Pine Avenue | Seattle          | Washington     | United States of America |
|        1800 | 90 Queen Street | Toronto          | Ontario        | Canada                   |
|        2500 | 8 Oxford Road  | Oxford            | Oxfordshire    | United Kingdom           |
+-------------+----------------+-------------------+----------------+--------------------------+
8 rows in set (0.00 sec)

mysql>
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MySQL 8.0 Command Line Cli

```
mysql> -- Q2: Write a query to find the name (first_name, last name), department ID and name of all the employees.
mysql> select e.first_name, e.last_name, e.department_id, d.department_name from employees e left join departments d on e.department_id = d.department_id;
+------------+-------------+---------------+-----------------+
| first_name | last_name   | department_id | department_name |
+------------+-------------+---------------+-----------------+
| Steven     | King        |            90 | Executive       |
| Neena      | Kochhar     |            90 | Executive       |
| Lex        | De Haan     |            90 | Executive       |
| Alexander  | Hunold      |            60 | IT              |
| Bruce      | Ernst       |            60 | IT              |
| David      | Austin      |            60 | IT              |
| Nancy      | Greenberg   |           100 | Finance         |
| Daniel     | Faviet      |           100 | Finance         |
| John       | Chen        |           100 | Finance         |
| Ismael     | Sciarra     |           100 | Finance         |
| Jose       | Urman       |           100 | Finance         |
| Matthew    | Weiss       |            50 | Shipping        |
| Adam       | Fripp       |            50 | Shipping        |
| Julia      | Nayer       |            50 | Shipping        |
| Irene      | Mikkilineni |            50 | Shipping        |
| James      | Landry      |            50 | Shipping        |
| Steven     | Markle      |            50 | Shipping        |
| John       | Russell     |            80 | Sales           |
| Karen      | Partners    |            80 | Sales           |
| Alberto    | Errazuriz   |            80 | Sales           |
| Peter      | Tucker      |            80 | Sales           |
| Allan      | McEwen      |            80 | Sales           |
| Clara      | Vishney     |            80 | Sales           |
| Sundar     | Ande        |            80 | Sales           |
| Ellen      | Abel        |            80 | Sales           |
| Kimberely  | Grant       |          NULL | NULL            |
| Jennifer   | Whalen      |            10 | Administration  |
| Michael    | Hartstein   |            20 | Marketing       |
| Pat        | Fay         |            20 | Marketing       |
| Shelley    | Higgins     |           110 | Accounting      |
| William    | Gietz       |           110 | Accounting      |
+------------+-------------+---------------+-----------------+
31 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q3: Write a query to find the name (first_name, last_name), job, department ID and name of the employees who works in London.
mysql> select e.first_name, e.last_name, e.job_id, e.department_id, d.department_name from employees e join departments d on e.department_id = d.department_
id join locations l on d.location_id = l.location_id where l.city = 'London';
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q4: Write a query to find the employee id, name (last_name) along with their manager_id and name (last_name).
mysql> select e.employee_id, e.last_name, e.manager_id, m.last_name as manager_last_name from employees e left join employees m on e.manager_id = m.employee
_id;
+-------------+-------------+------------+-------------------+
| employee_id | last_name   | manager_id | manager_last_name |
+-------------+-------------+------------+-------------------+
|         100 | King        |       NULL | NULL              |
|         101 | Kochhar     |        100 | King              |
|         102 | De Haan     |        100 | King              |
|         103 | Hunold      |        102 | De Haan           |
|         104 | Ernst       |        103 | Hunold            |
|         105 | Austin      |        103 | Hunold            |
|         108 | Greenberg   |        101 | Kochhar           |
|         109 | Faviet      |        108 | Greenberg         |
|         110 | Chen        |        108 | Greenberg         |
|         111 | Sciarra     |        108 | Greenberg         |
|         112 | Urman       |        108 | Greenberg         |
|         120 | Weiss       |        100 | King              |
|         121 | Fripp       |        100 | King              |
|         125 | Nayer       |        120 | Weiss             |
|         126 | Mikkilineni |        120 | Weiss             |
|         127 | Landry      |        121 | Fripp             |
|         128 | Markle      |        121 | Fripp             |
|         145 | Russell     |        100 | King              |
|         146 | Partners    |        100 | King              |
|         147 | Errazuriz   |        100 | King              |
|         150 | Tucker      |        145 | Russell           |
|         158 | McEwen      |        146 | Partners          |
|         162 | Vishney     |        147 | Errazuriz         |
|         166 | Ande        |        147 | Errazuriz         |
|         174 | Abel        |        147 | Errazuriz         |
|         178 | Grant       |        147 | Errazuriz         |
|         200 | Whalen      |        101 | Kochhar           |
|         201 | Hartstein   |        100 | King              |
|         202 | Fay         |        201 | Hartstein         |
|         205 | Higgins     |        101 | Kochhar           |
|         206 | Gietz       |        205 | Higgins           |
+-------------+-------------+------------+-------------------+
31 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q5: Write a query to find the name (first_name, last_name) and hire date of the employees who was hired after 'Jones'.
mysql> select first_name, last_name, hire_date from employees where hire_date > (select hire_date from employees where last_name = 'Jones');
Empty set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q6: Write a query to get the department name and number of employees in the department.
mysql> select d.department_name, count(e.employee_id) as no_of_employees from departments d left join employees e on d.department_id = e.department_id group
 by d.department_name;
+-----------------+-----------------+
| department_name | no_of_employees |
+-----------------+-----------------+
| Administration  |               1 |
| Marketing       |               2 |
| Shipping        |               6 |
| IT              |               3 |
| Sales           |               8 |
| Executive       |               3 |
| Finance         |               5 |
| Accounting      |               2 |
+-----------------+-----------------+
8 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q7: Write a query to find the employee ID, job title, number of days between ending date and starting date for all jobs in department 90.
mysql> select h.employee_id, j.job_title, datediff(h.end_date, h.start_date) as no_of_days from job_history h join jobs j on h.job_id = j.job_id where h.dep
artment_id = 90;
+-------------+-------------------------+------------+
| employee_id | job_title               | no_of_days |
+-------------+-------------------------+------------+
|         200 | Administration Assistant |       2100 |
+-------------+-------------------------+------------+
1 row in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

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mysql> -- Q8: Write a query to display the department ID and name and first name of manager.
mysql> select d.department_id, d.department_name, e.first_name as manager_first_name from departments d join employees e on d.manager_id = e.employee_id;
+---------------+-----------------+--------------------+
| department_id | department_name | manager_first_name |
+---------------+-----------------+--------------------+
|            10 | Administration  | Jennifer           |
|            20 | Marketing       | Michael            |
|            50 | Shipping        | Adam               |
|            60 | IT              | Alexander          |
|            80 | Sales           | John               |
|            90 | Executive       | Steven             |
|           100 | Finance         | Nancy              |
|           110 | Accounting      | Shelley            |
+---------------+-----------------+--------------------+
8 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q9: Write a query to display the department name, manager name, and city.
mysql> select d.department_name, concat(e.first_name, ' ', e.last_name) as manager_name, l.city from departments d join employees e on d.manager_id = e.empl
oyee_id join locations l on d.location_id = l.location_id;
+-----------------+-------------------+---------------------+
| department_name | manager_name      | city                |
+-----------------+-------------------+---------------------+
| IT              | Alexander Hunold  | Southlake           |
| Shipping        | Adam Fripp        | South San Francisco |
| Administration  | Jennifer Whalen   | Seattle             |
| Executive       | Steven King       | Seattle             |
| Finance         | Nancy Greenberg   | Seattle             |
| Accounting      | Shelley Higgins   | Seattle             |
| Marketing       | Michael Hartstein | Toronto             |
| Sales           | John Russell      | Oxford              |
+-----------------+-------------------+---------------------+
8 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q10: Write a query to display the job title and average salary of employees.
mysql> select j.job_title, avg(e.salary) as average_salary from employees e join jobs j on e.job_id = j.job_id group by j.job_title;
+------------------------------+----------------+
| job_title                    | average_salary |
+------------------------------+----------------+
| Public Accountant            |    8300.000000 |
| Accounting Manager           |   12008.000000 |
| Administration Assistant     |    4400.000000 |
| President                    |   24000.000000 |
| Administration Vice President |   17000.000000 |
| Accountant                   |    8175.000000 |
| Finance Manager              |   12008.000000 |
| Programmer                   |    6600.000000 |
| Marketing Manager            |   13000.000000 |
| Marketing Representative     |    6000.000000 |
| Sales Manager                |   13166.666667 |
| Sales Representative         |    8983.333333 |
| Stock Clerk                  |    2625.000000 |
| Stock Manager                |    8100.000000 |
+------------------------------+----------------+
14 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q11: Write a query to display job title, employee name, and the difference between salary of the employee and minimum salary for the job.
mysql> select j.job_title, concat(e.first_name, ' ', e.last_name) as employee_name, e.salary - j.min_salary as salary_difference from employees e join jobs
j on e.job_id = j.job_id;
+-----------------------------+------------------+-------------------+
| job_title                   | employee_name    | salary_difference |
+-----------------------------+------------------+-------------------+
| Public Accountant           | William Gietz    |           4100.00 |
| Accounting Manager          | Shelley Higgins  |           3808.00 |
| Administration Assistant    | Jennifer Whalen  |           1400.00 |
| President                   | Steven King      |           4000.00 |
| Administration Vice President | Neena Kochhar  |           2000.00 |
| Administration Vice President | Lex De Haan    |           2000.00 |
| Accountant                  | Daniel Faviet    |           4800.00 |
| Accountant                  | John Chen        |           4000.00 |
| Accountant                  | Ismael Sciarra   |           3500.00 |
| Accountant                  | Jose Urman       |           3600.00 |
| Finance Manager             | Nancy Greenberg  |           3808.00 |
| Programmer                  | Alexander Hunold |           5000.00 |
| Programmer                  | Bruce Ernst      |           2000.00 |
| Programmer                  | David Austin     |            800.00 |
| Marketing Manager           | Michael Hartstein|           4000.00 |
| Marketing Representative    | Pat Fay          |           2000.00 |
| Sales Manager               | John Russell     |           4000.00 |
| Sales Manager               | Karen Partners   |           3500.00 |
| Sales Manager               | Alberto Errazuriz|           2000.00 |
| Sales Representative        | Peter Tucker     |           4000.00 |
| Sales Representative        | Allan McEwen     |           3000.00 |
| Sales Representative        | Clara Vishney    |           4500.00 |
| Sales Representative        | Sundar Ande      |            400.00 |
| Sales Representative        | Ellen Abel       |           5000.00 |
| Sales Representative        | Kimberely Grant  |           1000.00 |
| Stock Clerk                 | Julia Nayer      |           1200.00 |
| Stock Clerk                 | Irene Mikkilineni|            700.00 |
| Stock Clerk                 | James Landry     |            400.00 |
| Stock Clerk                 | Steven Markle    |            200.00 |
| Stock Manager               | Matthew Weiss    |           2500.00 |
| Stock Manager               | Adam Fripp       |           2700.00 |
+-----------------------------+------------------+-------------------+
31 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q12: Write a query to display the job history that were done by any employee who is currently drawing more than 10000 of salary.
mysql> select * from job_history where employee_id in (select employee_id from employees where salary > 10000);
+-------------+------------+------------+------------+---------------+
| employee_id | start_date | end_date   | job_id     | department_id |
+-------------+------------+------------+------------+---------------+
|         101 | 1997-09-21 | 2001-10-27 | AC_ACCOUNT |           110 |
|         101 | 2001-10-28 | 2005-03-15 | AC_MGR     |           110 |
|         102 | 2001-01-13 | 2006-07-24 | IT_PROG    |            60 |
+-------------+------------+------------+------------+---------------+
3 rows in set (0.00 sec)

mysql>
```

MySQL 8.0 Command Line Cli

```
mysql> -- Q13: Write a query to display department name, name (first_name, last_name), hire date, salary of the manager for all managers whose experience is
 more than 15 years.
mysql> select d.department_name, e.first_name, e.last_name, e.hire_date, e.salary from departments d join employees e on d.manager_id = e.employee_id where
timestampdiff(year, e.hire_date, curdate()) > 15;
+-----------------+------------+-----------+------------+----------+
| department_name | first_name | last_name | hire_date  | salary   |
+-----------------+------------+-----------+------------+----------+
| Administration  | Jennifer   | Whalen    | 1995-09-17 |  4400.00 |
| Marketing       | Michael    | Hartstein | 2004-02-17 | 13000.00 |
| Shipping        | Adam       | Fripp     | 2005-04-10 |  8200.00 |
| IT              | Alexander  | Hunold    | 2006-01-03 |  9000.00 |
| Sales           | John       | Russell   | 2004-10-01 | 14000.00 |
| Executive       | Steven     | King      | 2003-06-17 | 24000.00 |
| Finance         | Nancy      | Greenberg | 2002-08-17 | 12008.00 |
| Accounting      | Shelley    | Higgins   | 2002-06-07 | 12008.00 |
+-----------------+------------+-----------+------------+----------+
8 rows in set (0.00 sec)

mysql>
```